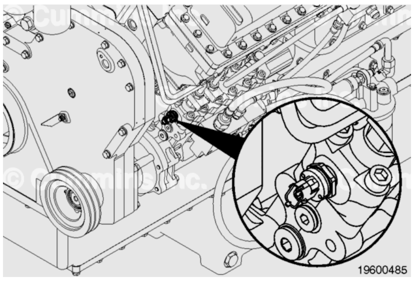
Trainings

General Information

The fuel rail pressure sensor is located on the front of the fuel pump.



Preparatory Steps

Note : The fuel system pressure MUST be relieved before removing the rail pressure sensor.

- Clean the engine. Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 000-999 in Section F. (/qs3/pubsys2/xml/en/procedures/28/28-000-999.html)
- Use contact cleaner, Part Number 3824510 or equivalent, to thoroughly clean all fittings and components before removal from the engine. Clean the connector fittings to remove as much debris as possible. It is very important that extra care is taken to keep the fuel connections clean during removal and installation. Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 204-008 in Section i. (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)
- Make sure that debris, water, steam, or cleaning solution does **not** enter the fuel system.

Remove

⚠ WARNING ⚠

The fuel pump, high-pressure fuel lines, and injector supply lines contain very high-pressure fuel. To reduce the possibility of injury and property damage, do not loosen any fittings while the engine is running.

⚠ WARNING ⚠

Some contingency fuel, such as a jet fuel and kerosene, are much more flammable than normal diesel fuel. Use extreme care to keep cigarettes, flames, pilot lights, sparks, arcing equipment and switches, and other sources of ignition away from and out of areas sharing ventilation.

⚠ WARNING ⚠

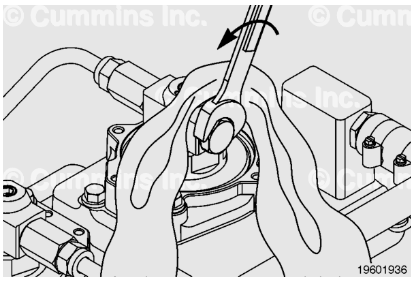
Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs on the fuel system, to reduce the possibility of fire and resulting severe personal injury, death, or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

⚠ WARNING ⚠

Pressure within the high-pressure fuel system must never be measured using a mechanical gauge. Fuel pressure values of over 1724 bar [25,000 psi] are possible. Under high pressure, a mechanical gauge can fail causing a high-pressure fuel leak resulting in personal injury and property damage.

⚠ CAUTION ⚠

A small amount of dirt and debris can be very harmful to the injectors and the cone seats on the injector high-pressure supply connections. Extra care is required to keep the fuel connections clean during removal and installation. Use the following procedure in the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528. Refer to Procedure 204-008 in Section i. (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)



⚠ CAUTION ⚠

Do not over-tighten or under-tighten the injector supply line fittings. A fuel leak can occur.

⚠ CAUTION ⚠

Do not spill or drain fuel into the bilge area when disconnecting or removing fuel lines, replacing filters, and priming the fuel system. Do not drop or throw filter elements into the bilge area. The fuel and fuel filters must be disposed of in accordance with local environmental regulations.

Note : The fuel system pressure MUST be relieved before removing the rail pressure sensor.

Cover the plug with a lint-free cloth. Slowly loosen the plug on the last injector of the high-pressure fuel lines 1/4-1/2 turn. The plug does **not** need to be removed to relieve the pressure. **ONLY** one plug needs to be removed.

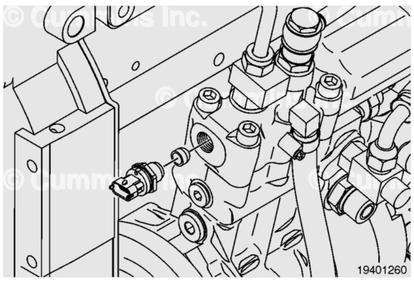
Connect INSITE™ electronic service tool and verify the fuel pressure has bled down by monitoring the fuel pressure.

Remove the plug and discard the o-ring.

Make sure the area around the fuel rail pressure sensor is clean.

Disconnect the fuel rail pressure sensor connector from the engine harness.

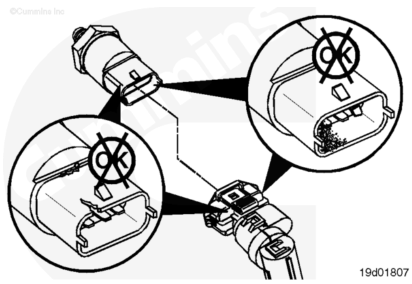
Remove the fuel rail pressure sensor and the sealing washer.



Inspect for Reuse

Inspect the engine harness connector and the fuel rail pressure sensor for the following:

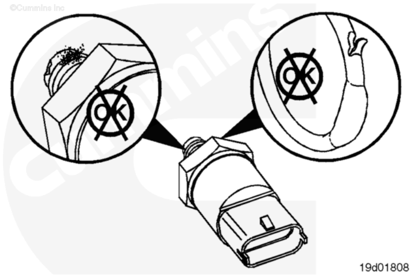
- Cracked or broken connector shell
- Missing or damaged connector seals
- Dirt, debris, or moisture in or on the connector pins
- Corroded, bent, broken, pushed back, or expanded pins.



Inspect the fuel rail pressure sensor for the following:

- Debris on the tip of the sensor
- Thread damage.

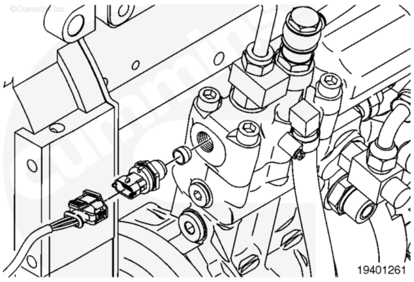
Replace the sensor if damage is found.



Install

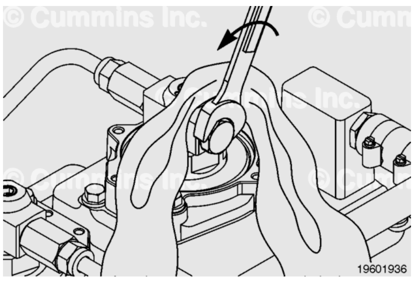
Install a new sealing washer, then install the injector metering fuel rail pressure sensor.

Torque Value: 136 n•m [100 ft-lb]



Install the plug, with a new o-ring, into the last injector connection.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

- Connect the engine harness to the fuel rail pressure sensor.
- Start the engine and check for leaks.

Last Modified: 08-Dec-2011